

Co-Drive: Experiencing Social Virtual Travel on a Car Trip

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ABSTRACT

The Co-Drive interactive experience at CHI2021 gives conference attendees the possibility to experience social virtual travelling by car either as the driver or the remote passenger. Through the dislocation of two prototypes in two different parts of the world, Co-Drive trips will be available in two different locations, one of which crowd-sourced among prospective CHI attendees. After experiencing the Co-Drive trip, participants will be able to share their experience in a collective way through a virtual meeting held during the conference and subsequently in an individual way through a phone call or video/audio recording from their car to the main author.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**; *Interaction design process and methods*; *Participatory design*;

KEYWORDS

remote travelling, social virtual reality

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1 CO-DRIVE AS A SERVICE FOR SOCIAL VIRTUAL TRAVEL

Co-Drive is a service concept that allows social virtual travelling by car between a driver of a vehicle and a remote passenger connected via virtual reality from home [1]. The concept is inspired by emergent automated car technologies, such as teleoperated driving, in an attempt to push their development beyond driving and safety functionality and draw in the design of novel mobility services into specific social context and situated everyday lives. The Co-Drive

concept enables novel social interactions between a driver and a remote passenger. It also aims to foster new social interactions, such as inter-generational encounters, for elderly remote passengers with reduced mobility and travel possibilities. Drawing from the concept of "Driver-Car assemblage" as a distinct social being acting in our society [7], the researchers look at Co-Drive as the new assemblage of "Driver-Remote Passenger-Automated Car" and probe the emergence of social "actions" through experience prototyping and co-creation with potential remote passengers and drivers [2] [6]. Co-Drive is envisioned as a service which unfolds throughout several stages and touchpoints until the remote passenger and the driver eventually meet at a Co-Drive stop and they depart together on a shared car trip [1]. According to the envisioned scenario, people who wish to engage in a remote Co-Drive car trip can select among the many stops around the world which have been featured in the growing Co-Drive Atlas [3] and place their avatar there to book a Co-Drive trip. A Co-Drive stop is the physical location where a Co-Drive trip can start from. A driver passing nearby the stop where an avatar has been placed would spot such an avatar as an AR visualization on the car windshield and can decide to pull over [5], start engaging remotely with the person embodied in the avatar and eventually board her/him in the car as a remote passenger for a shared trip together. The car would be equipped with the technology enabling the remote social travel experience, consisting of a remote passenger pod mounted on the headrest of the passenger seat [Fig.1], which would enable video live streaming from the car to the remote passenger's VR headset and bilateral audio communication from the driver to the remote passenger and back.

2 THE CO-DRIVE INTERACTIVE EXPERIENCE

The Co-Drive interactive experience at CHI 2021 invites people to board a Co-Drive trip either as a driver or as a remote passenger. Due to the limited number of available Co-Drive pods (2), only a driver will be recruited among prospective CHI attendees well in advance, so to allow for the shipping of the pod prototype to her/his premises. The other interested attendees will be able to book a Co-Drive trip over the six days of the conference, following a schedule which takes care of drivers availability. It is envisioned that up to 5 trips a day will be running in each one of the two distinct locations serving as Co-Drive stops, so that a maximum of around sixty attendees could experience turning into a Co-Drive

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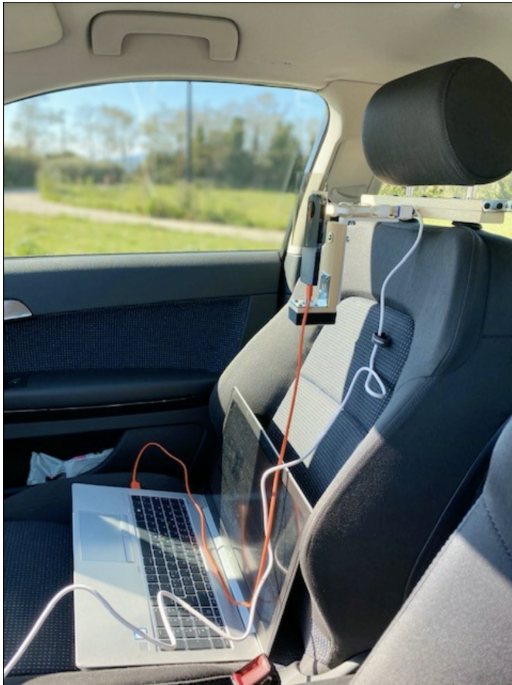


Figure 1: The Co-Drive Pod early prototype

remote passenger. A waiting list will be created for all the other prospective remote passengers who didn't make the available slots, as well as a streaming channel, mirroring the remote passengers experience, will be made available on the project website [4], so that the rest of the audience could still get involved even if not experiencing it in first person.

2.1 The trip locations

Two trip locations will be staged as Co-Drive stops: the first one will be in Rome, Italy, where the author of the project lives and will be able to drive her own car even under the unfortunate scenario of major Covid-19 restrictions in Rome; the second one will be identified among all the Co-Drive stop submissions which would be collected through the public call for the CHI2021 Co-Drive Experience[3]. Preference will be granted to submission from prospective drivers who would give plenty of availability to driving and hosting remote passengers during the days of the conference and who are located in countries other than Italy.

2.2 The participants and trips schedule

The recruitment of the prospective driver and remote passengers will follow two different and asynchronous processes. The recruitment of the driver will run first and only after that, the remote passengers recruitment will start. A public call for the driver will be released and spread on the CHI2021 social channels in the second half of February, as well as through the Ubicomp mailing list, and will stay open for a month. prospective drivers will be required to fill the Co-Drive stops Google form for CHI2021 (similar to the one related to the Co-Drive Atlas [3]) [Fig.3]. The selection of the

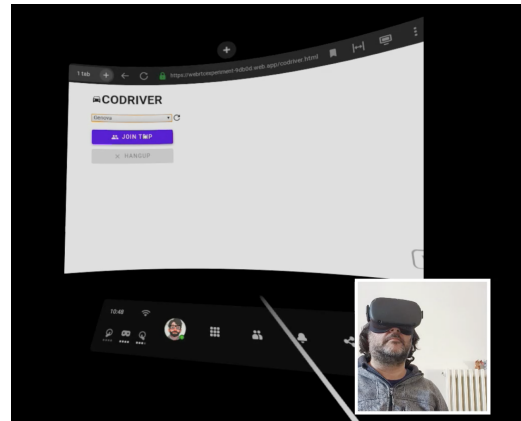


Figure 2: A remote passenger joining a trip through a VR headset

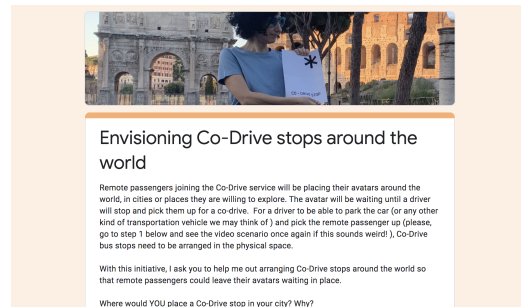


Figure 3: The Co-Drive stop public call

driver will be done according to the availability to drive, given that preference will be granted to drivers located outside of Italy. Once the driver would be chosen, in the second half of March the Co-Drive pod will be shipped to her/him. Through remote calls, the driver will be guided on how to mount the equipment on her/his car and get the pod working. A schedule of 20 minutej-trip slots will be created with the driver, hoping that all the six conference days could be covered.

The prospective remote passengers will be able to book their trip on the project website (the link will be released through CHI2021 social channels and Ubicomp mailing list) starting from April. To experience the Co-Drive trip, they will not strictly be required to have a VR headset since they will be able to join either on screen or headset when accessing the trip from the provided link [Fig.2]. Remote passengers will be required to choose a favourite destination, but the organizers may operate changes to match the language/s spoken by the driver. Eventually a waiting will be created.

The rest of the audience could still get involved by watching the Co-Drive trips which get live streamed on the project website[4].

3 WRAPPING UP THE EXPERIENCE & CONCLUSION

To wrap up the experience, the author of the project will organize a virtual meeting on Mozilla Hubs on the last day of the conference with all the participants who engaged with the Co-Drive Experience at CHI 2021. Thoughts and suggestions from people will be collected in an informal way. This meeting is far away from being a focus group, but a casual meeting where people can attend, leave and participate at their own pace. Moreover, when the conference would be over, participants will be asked to reflect again individually on their own Co-Drive experience and connect again with the author of the project, either making a traditional phone call to her or recording a video/audio message to send to her. They will be required to perform this last bit from their car, so to keep the intimacy and informal context which a car trip grants. The Co-Drive interactive experience at CHI2021 will contribute to the Co-Drive on-going research, helping to answer questions like: "What would it be like to step into a foreigner's car as a remote passenger? What would it be like to host a remote passenger in the own car? What

would the driver and remote passenger talk about? Why would they like or dislike traveling like this?"

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